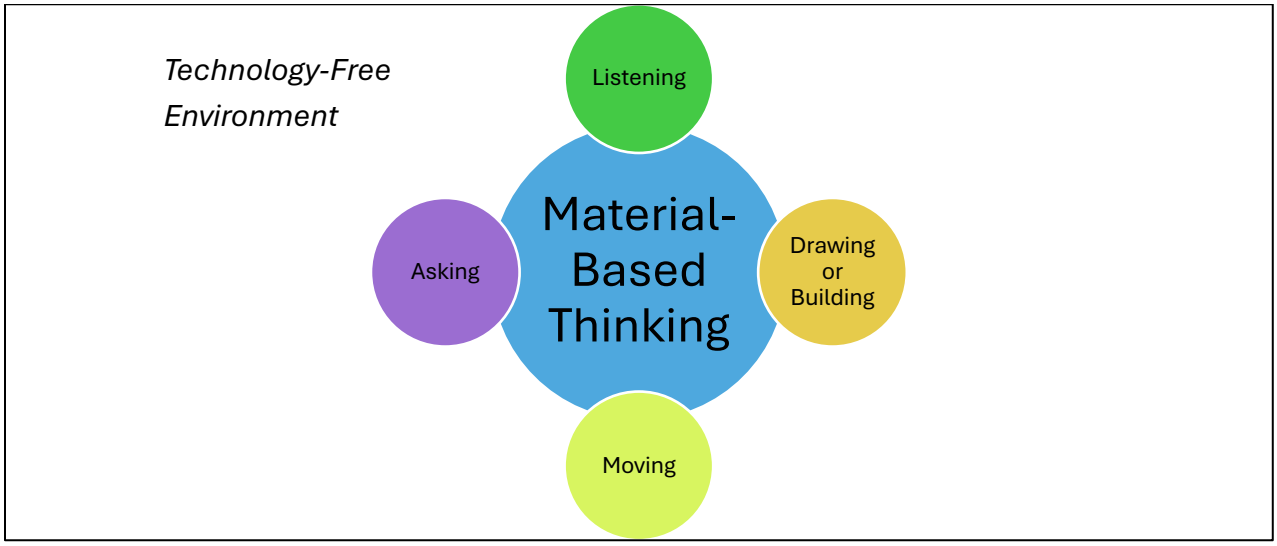
Material-Based Thinking (MBT) Framework for Business Education

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Material-Based Thinking (MBT) is a critical part of the learning process, in classrooms and in the field. Understanding is enhanced and outcomes improve when MBT is part of the process. It is best done in technology-free spaces but can be done whenever humans have technology-free periods of time. It is meant to be done in sequence with technology/AI use, either before, after, or in-between.

MBT involves 4 actions:

- **Asking:** Asking questions of teammates, customers, or other humans who have relevant information and experiences to the problem or question at hand. It means preferring open questions to ones that invite a yes or no, and treating the first answer as the beginning rather than the end of an exchange. Good asking is specific, unhurried, and willing to look uninformed in front of someone who knows more.
- **Listening:** Listening to the information and experiences shared by others. It includes letting an answer change the question that follows, noticing what a person does not say or declines to say, and recording their words rather than a paraphrase already shaped toward a conclusion. Notes taken by hand help here, because the slower pace forces selection and keeps attention on the speaker.

- **Drawing or Building:** Using human brains and hands to draw non-digital diagrams, make lists or tables, or build prototypes from available materials — paper, whiteboards, sticky notes, index cards, cardboard, tape, string, blocks, and the actual objects under study. The output might be a process map, a floor layout, a decision table, a sketch of an interface, or a working physical mock-up.
- **Moving:** Moving the body through physical spaces, both to collect information through observation and to reflect on current thinking. This includes site visits, tracing a single unit from one end of a process to the other, or walking meetings where a problem is discussed away from a table and a screen.